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High rate of muscle injury despite no changes in physical, physiological and psychophysiological parameters in a professional football team during a long-congested fixture period

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ABSTRACT

This study aimed to analyse match availability, participation, and muscle injury incidence and to compare the effect of time intervals between matches (3–4 versus 6–7 days) on physical, physiological, and psychophysiological parameters in a professional football team during a prolonged congested fixture period. Thirteen professional male football players (29.2 ± 4.8 years old; 78.5 ± 8.3 kg; 179.3 ± 8.8 cm;) participated. Data were collected during 17 consecutive weeks for 35 official matches separated by an average interval of ≤ 3.5 days. Results showed a player availability of 84.8 ± 16.1% while match participation was 68.8% ± 6.4%. Muscle injury incidence was 19.0/ 1,000 hours of match exposure. These injuries occurred after individual players participated in sequences of 8.3 ± 3.3 consecutive matches with intervals of 3.8 ± 0.8 days. No differences were observed in physical performance or in fatigue-related parameters irrespective of the time interval between matches . a high player availability rate was observed. No differences were observed in physical, physiological, and psychophysiological indices of performance when comparing shorter and longer time intervals between consecutive matches.Prolonged exposure to fixture congestion resulted in an high risk of incurring muscle injury.

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Electronic performance tracking systems; soccer; match congestion; physiological parameters; injury rate

Introduction

In elite football, teams regularly participate in two matches per week, often over extended periods (Bengtsson et al., 2013). Teams can also play up to 3 matches in 9 days and participate in more than 65 official matches, in multiple competitions, over an 11-month

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season, equating to around 6 matches per month (Ernst & Young, 2019). Commonly referred to as fixture congestion, these periods require matches to be played successively separated by short intervals of only 3 or 4 days (Bengtsson et al., 2018; Howle et al., 2020). Short congested match cycles, containing sequences of two to three consecutive matches, are common (Mohr et al., 2016) while some teams can frequently encounter longer congested sequences of 5 to 7 consecutive matches (Carling et al., 2015; Dellal et al., 2015).

To ensure that players are not continually exposed to fixture congestion coaching staff can employ rotation strategies (Carling et al., 2015). Varley et al. (2018) investigated participation rates in elite football players and found that when teams had assured progression in a tournament, coaching policy was to rest key players. Carling et al. (2015) revealed a trend for a sharp decline in participation as the number of matches in the congested cycles increased. Contextual factors will also affect team selection decisions. These include the importance of each match, quality of opposition, injuries, and suspensions.

Regarding physical performance indices, match congestion is generally shown not to impact overall physical performance represented by the total distance covered match-to-match (Julian et al., 2020). However, while professional players are generally shown to cope "physically" over short periods of fixture congestion (Carling et al., 2015a), research suggests there is a considerable increase in the incidence of muscle injuries (Bengtsson et al., 2018; Gouttebarge et al., 2019). A substantial increase exists for the overall risk of incurring injury and soft-tissue injury specifically compared to a 1-match per week schedule (Bengtsson et al., 2018; Carling et al., 2015; Howle et al., 2020). Research shows that 6–10-day intervals separating successive match exposures result in muscle injury rates approximately 20% lower than those following a ≤ 3 days interval (Bengtsson et al., 2018).

Following a football match, physiological markers of muscle damage, inflammatory responses and reduced physical performance in testing can remain for up to 72 hours (Mohr et al., 2016). Increases in perceived soreness, fatigue (Mohr et al., 2016; Rampinini et al., 2011) and high levels of CK (Nédélec, 2014) have also been reported. Generally, evidence shows that the magnitude of change in these disturbances begins within the first 24 hours following match-play with values peaking between 24 and 48 hours whilst potentially remaining affected for up to 4 days to pre-match values (Ispirlidis et al., 2008). As such, some players might be participating in competition when not fully recovered physically, physiologically, or mentally. To our knowledge, no research has verified changes in these variables over a single long uninterrupted congested period (e.g. ≥ 10 consecutive weeks with intervals between matches ≤ 4 days) that contemporary football clubs can face.

Extensive information regarding the acute impact of congested fixture periods has been provided in the last decade (Bengtsson et al., 2018; Carling et al., 2012, 2015). However, to the best of our knowledge, no other study has evaluated physical performance and injury over an extended period of congestion (>8 matches). In addition, it is not known whether an individual extension in the recovery time between matches could result in improved recovery and consequently maintain or improve better physical performance in the next match across a long uninterrupted congested period. Therefore, the aims of this study in a reference elite professional football team were two-

fold: 1) to analyse match availability, participation, and muscle injury incidence during a prolonged congested fixture period, 2) to compare the effect of time intervals separating matches on physical, physiological, and psychophysiological parameters during this period.

Methods

Sample

A total of 13 male professional football players (29.2 ± 4.8 years, 78.5 ± 8.3 kg, 179.3 ± 8.8 cm, body fat 10.5 ± 1.7%, YoYo Intermittent Recovery Test – Level1- YIRT1: 1.820 ± 343 metres) participated. These included: central defenders (3), fullbacks (2), midfielders (4), and forwards (4). Goalkeepers were excluded from the analysis. The players belonged to the squad of a first division Brazilian football team and had professional experience of at least two years in training and competition at national and international levels. All players were submitted to medical evaluations by the club's medical staff and presented adequate health status for the practice of professional football.

Ethical standards

The study was approved by the local Ethics and Research Committee and respected all the standards established by the National Health Council (Res. 466/2012). All volunteers signed the Free and Informed Consent Form. The data were only included following consent obtained from the participants. A consent letter was also obtained from the club. The anonymity of the participants was preserved throughout the process. This study respected all the standards established by the Declaration of Helsinki.

Experimental design

Data collection was conducted during a continuous competitive period of 17 weeks. During this period, the team participated in 35 consecutive official matches, with an interval of ≤ 4 days. The team participated in 4 different competitions at national and international level. This period is referred to as the Congested Match Period (CFP). The official matches were played during midweek (Wednesday or Thursday) and on weekends (Saturday or Sunday) at the following times: 4 pm, 6 pm, 7:30 pm, or 9 pm. During the competitive season, there were no changes in technical and medical staff.

Figure 1 shows the sequential distribution of weeks from the start of the season to the end of the data collection period. The number of matches played and the field training sessions performed are reported for each week.

Players' participation in matches was decided by the coaching staff, as well as influenced by suspensions (e.g. yellow cards), and occurrence of injuries. Thus, even if the team played consecutive matches separated by an interval of <4 days, players might not participate for the reasons mentioned above and could therefore have had an individual

Month	January			February				March					April				May					June				July				August	
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
N. Matches played				1	2	2	1	1	2	1	2	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
N. Field training sessions	5	6	5	4	3	4	5	4	3	4	3	5	4	5	2	3	2	1	1	2	1	2	1	1	2	2	2	2	3	2	2
Period	A			B													C														

Figure 1. Sequential distribution of weeks throughout the sports season. Legend: A) Pre-season (3 weeks); B) Initial period of competitive calendar: 15 matches in 11 weeks; C) Prolonged Congested fixture period: 2 matches per week over 17 consecutive weeks. The data collection was conducted in this period.

Table 1. Player participation and availability metrics.

Metric	Definition
Absolute Availability (AA)	Number of matches in which the player was available to play in the analysed period
Relative Availability (RA%)	Percentage of matches the players were available to play in this period, according to the total number of matches played by the team in the same period
Absolute Participation (AP)	Number of matches in which each player participated in this period (regardless of the time of participation in the match)
Relative Participation (RP%)	Percentage of matches that the player participated in the period (regardless of the time of participation in the match), according to the total number of matches played by the team in the same period.
Absolute participation in 90 minutes of play (P90N)	Number of complete matches (90 min) in which each player participated during this period.
Relative participation in 90 minutes of play (P90%)	Percentage of complete matches (90 min) that the player has participated, according to the total number of matches he has participated (regardless of the time of participation) in the same period
Absolute participation in ≥ 75 minutes match (P75N)	Number of matches in which each player participated for ≥75 minutes of the match in this period.
Relative participation in ≥ 75 minutes match (P75%)	Percentage of matches that the player participated for ≥ 75 minutes of the matches, depending on the total number of matches that he participated (regardless of the time of participation) in the same period.

interval of up to 7 days between matches. These two different individual intervals between matches (3–4 and 6–7 days) were used to compare metrics relating to physical performance and fatigue.

Match participation

Player participation and availability were analysed using metrics based upon availability and participation (Carling et al., 2015) and are presented in Table 1. Individual analyzes were performed to quantify the effective participation of each player in the congested period. During the prolonged congested fixture period (PCFP) 35 consecutive matches were played with an average interval of 3.5 days.

Muscle injury incidence

All muscle injuries resulting from a football match during this period were prospectively diagnosed and registered by the same club's sports physician, using magnetic resonance imaging. The total time (in days) of absence from training activities and matches owing to

the occurrence of these injuries was recorded individually. A muscle injury was defined as a traumatic distraction or overuse injury, leading to the player being unable to take full part in future football training or match play (Bengtsson et al., 2013). The injury incidence was calculated for 1,000 hours of exposure to match-play (Bengtsson et al., 2013; Dellal & Wong, 2013).

Match-play physical performance

Physical performance in matches was monitored using a semi-automatic optical player tracking system via digital footage collected by a set of fixed cameras (InStat®, Moscow, Russia). Optical tracking systems are used ubiquitously in studies of physical performance in elite football match-play (Carling et al., 2012, 2015; Dellal & Wong, 2013). The InStat system has recently been independently approved by the FIFA Quality Programme for Electronic & Performance Tracking Systems (FIFA Quality Performance Report, 2020).

The variables used to characterize physical performance were total distance covered, distances covered between 14.4 km/h and 20.0 km/h, 20.1 km/h and 25.0 km/h, and above 25.0 km/h. Data are presented minute by minute and during effective playing time as this factor is shown to impact match-to-match running performance (Dellal et al., 2015).

The criteria for inclusion in the analysis of physical performance indices were: minimum participation of 75 minutes in matches (Dellal et al., 2015); and the player should have participated in a previous match within the congested period with an interval of ≤ 7 days. In situations where the player participated in less than 75 min of a match, the data were not included for analyses of physical performance. However, these data were counted as participation within the congested period of matches. Players absent for >21 days were excluded from the analysis.

Physiological analysis

On the day after the matches (16 to 21 hours), serum CK concentration dosage assessments were performed. To avoid interference from any activity other than the previous day's match, these procedures were performed 1 hour before the following training session. The measurement was performed by taking a capillary blood sample extracted from the player's digital pulp through a small perforation made with the aid of a single, automatically driven disposable lancet (Accu-Chek Safe T Pro Plus; Roche, Switzerland). The blood sample was collected using a heparinized glass capillary (Capilette®, Selzer, Germany) and placed in contact with a reagent strip for CK (Roche Diagnostics Gmbh, A, Germany) that was inserted into the mini portable biochemical analysis laboratory (Reflotron Analyser, Boehringer Mannheim), as performed in previous research (Coelho et al., 2016).

Psychophysiological parameters of fatigue

To estimate the fatigue perceived by the players following the participation in the match held the previous day, the players answered two questionnaires using an ordinal classification scale. To estimate the players' subjective perception of recovery (SPRec), a 6–20 Total quality recovery scale (Kenttä & Hassmén, 1998) was employed. The perception of delayed onset muscle soreness (DOMS) was measured using the scale proposed by Vickers (2001).

Statistical analysis

Results are presented as means and standard deviations. Kolmogorov Smirnov's test was performed to verify data normality. The paired Student's T-test was performed to compare physical performance, (SPRec) and (DOMS) between matches played after "3–4D interval" and "6–7D interval". The effect size was calculated using the Cohen test (d) to identify the magnitude of the differences in means (Cohen, 1992). The effect size values were classified as small (< 0.2), medium (0.2–0.79), and large (≥ 0.8). The level of statistical significance adopted was α = 0.05. All data were analysed using the Statistical Package for the Social Sciences (IBM Corp. Released 2016. IBM SPSS Statistics for Windows, Version 24.0. Armonk, NY: IBM Corp.) software.

Results

Match participation during the CFP

During the CFP, the team participated in 35 matches, with an average interval of 3.5 ± 0.7 days between successive matches. Matches were mainly separated by individual intervals of 3–4 days (76.6 ± 9.9% of the total) in relation to participation in the previous match. Information on player availability and participation is presented in Table 2.

For the 13 players involved in this analysis, 4 participated in over 80% of the 35 matches while 2 players participated in 31 matches (88.6%). No player participated in all 35 matches. Collectively, these players recorded a total of 313 individual match participations of which 76.6% ± 9.9% occurred following a ≤ 4 days interval after the previous match. All 13 players had at least one personal sequence ≥ of 6 consecutive matches with 3–4 days intervals. Individual sequences are presented in Table 3.

Incidence of muscular injuries

A total of 11 muscle injuries were observed during the 35-match sequence equating to an incidence rate of 19.0 muscle injuries for 1,000 hours of match exposure. These 11 muscle injuries were sustained by 7 of the 13 players. The average absence for muscle injuries was 19.4 ± 10.3 days. The injuries were situated in the following muscles: hamstring (6, with 1 recurrence), calf (3, with 1 recurrence), adductor (1), rectus femoralis (1). These injuries occurred on average following an individual sequence of 8.3 ± 3.3 consecutive matches with intervals of 3.8 ± 0.8 days between matches. Based on the Gaussian distribution, in

Table 2. Availability and participation in 35 official matches (n = 13) during the congested fixture period (17 weeks).

Metric	Values
Absolute Availability (AA)	29.7 ± 4.8
Relative Availability (RA%)	84.8 ± 16.1
Absolute Participation (AP)	24.1 ± 5.7
Relative Participation (RP%)	68.8 ± 6.4
Absolute participation in 90 minutes of play (P90N)	14.1 ± 8.1
Relative participation in 90 minutes of play (P90%)	59.7 ± 29.8
Absolute participation in ≥ 75 minutes match (P75N)	16.0 ± 7.7
Relative participation in ≥ 75 minutes match (P75%)	67.4 ± 26

practical terms, this represents an increased risk of ~68% for the occurrence of muscular injuries happening with individual sequences of 5.0 to 11.6 matches with a recovery interval between these of 3.0 to 4.6 days.

Physical, physiological, and psychophysiological variables

The physical performance measured by the total distance covered (normalized per minute of activity) and the distance covered (normalized per minute of activity) in the highest intensity ranges (14.4 to 20.0 km/h: $p = 0.136$; 20.0 to 25.0 km/h: $p = 0.682$; above 25.0 km/h: $p = 0.371$) did not present differences when comparing the two different time intervals between matches (3–4D vs 6–7D). Similarly, no differences were observed for the fatigue-related parameters: CK ($p = 0.443$); SPRec ($p = 0.424$); DOMS ($p = 0.701$). Data are presented in Table 4.

Discussion

The aims of this study in a reference elite professional football team were two-fold: 1) to analyse match availability, participation, and muscle injury incidence during a prolonged congested fixture period; 2) compare the effect of short versus longer time intervals (3–4 or 6–7 days) between successive matches on physical, physiological, and psychophysiological variables during this period. The main findings were 1) a high average percentage of player availability (84.8 ± 16.1%), 2) a muscle injury rate of 19.0 per 1,000 hours of match exposure, 3) no difference for physical, physiological, and psychophysiological indices when comparing values for shorter and longer time intervals between consecutive matches.

In the current study, the club participated in a prolonged CFP lasting 17 consecutive weeks. To our knowledge, no other report has examined fixture congestion over a similar duration. Player availability was approximately 85% over the period. Research has shown that player availability is important for team success as higher match availability is associated with final league ranking and increased points won per league match in elite football (Hägglund et al., 2013). In contrast, increased player unavailability can contribute to a relatively weaker line-up and higher match intensities for the starting players (Windt et al., 2018). When the effective participation in the 35 matches of the CFP was evaluated (regardless of actual time played in each match), the 13 players participated in approximately 70% of all matches. The highest uninterrupted individual participation in consecutive matches with an interval ≤ 4 days in this study was 11 matches (performed by a single player). There were 21 individual participations in ≥ 6 successive matches with intervals ≤ of 4 days, 8 consecutive matches being the individual sequence that most frequently occurred during the entire CFP. Thus, the players participated extensively in the matches during this period and notably in relatively long consecutive individual sequences of matches. Even when national and international matches were accumulated at the same time, the main players were not spared from exposure to long sequences of matches separated by short intervals.

Research has demonstrated a higher incidence rate of non-contact injuries during congested versus non-congested periods (Bengtsson et al., 2013; Dellal et al., 2015). Here, 7 of the 13 players sustained muscle injuries (n = 11). Previous data collected from UEFA Elite Clubs over an eleven year period (Bengtsson et al., 2013) presented an average muscle injury rate of 11.9/1,000 hours of match exposure for league matches and 13.1 for UEFA Champions League matches played after ≤ 4-day intervals. These rates were

Table 3. Number of individual sequences of participation in ≥ 6 consecutive matches played with a 3–4-day interval. Distribution of participation in matches during the congested period.

	Number of matches played in the congested period	Number of individual sequences of >6 consecutive matches	Match 1	Match 2	Match 3	Match 4	Match 5	Match 6	Match 7	Match 8	Match 9	Match 10	Match 11	Match 12	Match 13	Match 14	Match 15	Match 16	Match 17	Match 18	Match 19	Match 20	Match 21	Match 22	Match 23	Match 24	Match 25	Match 26	Match 27	Match 28	Match 29	Match 30	Match 31	Match 32	Match 33	Match 34	Match 35
Player 1	19	2					■	■	■	■	■	■	■	■	■										■		■		■	■		■	■	■	■	■	■
Player 2	30	3	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■				■	■		■	■	■	■	■	■	■	■	■	■	■		■	■
Player 3	31	3	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	■	■	■		■	■	■	■	■	■	■		■	■	■		■	■
Player 4	31	1	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■	■		■	■	■	■	■		■	■	■	■	■		■	■
Player 5	14	1				■			■		■	■	■	■	■	■	■		■	■	■	■															■
Player 6	24	1	■		■		■	■		■	■	■	■	■	■	■	■	■	■			■		■	■	■	■	■	■	■	■					■	
Player 7	26	2	■	■	■	■	■	■		■	■	■	■	■	■	■	■	■						■	■	■		■	■	■		■	■	■		■	■
Player 8	17	1	■	■	■	■	■	■	■	■										■	■	■		■	■								■	■		■	■
Player 9	21	1	■	■	■	■	■	■	■	■		■	■	■	■	■											■	■		■		■	■	■		■	■
Player 10	27	2	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		■	■		■	■	■		■					■	■		■	■	■	■
Player 11	26	1	■	■	■	■	■		■		■		■	■	■		■	■		■	■		■	■	■	■	■	■		■	■	■	■	■			■
Player 12	29	3	■	■	■	■	■	■		■	■	■		■	■	■	■	■	■	■	■	■		■	■	■	■	■	■		■	■		■	■	■	
Player 13	18	1							■			■				■	■	■	■	■	■		■	■	■	■	■	■			■	■	■		■		■

Table 4. Comparison of physical performance and fatigue parameters between matches played after 3–4 days vs 6–7 days intervals (*All distances are presented as metres covered per minute of playing time).

Variables	3–4 day interval	6–7 day interval	Effect Size	P value
Total distance covered (m/min)	113.69 (± 8.19)	115.53 (± 7.85)	−0.22	0.267
Distance covered at 14.4 to 20.0 km/h*	18.91 (± 3.17)	20.03 (± 3.08)	−0.34	0.136
Distance covered at 20.1 to 25.0 km/h*	7.96 (± 2.12)	8.14 (± 2.12)	−0.08	0.682
Distance covered above 25.0 km/h*	1.39 (±0.68)	1.24 (±0.89)	−0.19	0.371
CK	689 (± 255)	665 (± 236)	0.09	0.443
SPRec	11.85 (± 0.87)	12.00 (± 1.26)	−0.12	0.424
DOMS	2.45 (± 0.54)	2.34 (±0.64)	0.16	0.701

significantly higher compared to matches played ≥ 6-day intervals. In the present study, the muscle injury rate during a long-congested period was 19.0/1,000 hours match exposure, suggesting that a prolonged congested fixture can result in an even greater risk of muscle injury incidence. A reasonable explanation for the higher match muscle injury rates, and notably in the hamstring muscle group reported during periods of match congestion, is the residual muscular fatigue post-match (Bengtsson et al., 2018) mainly due to insufficient recovery time (Bengtsson et al., 2017; Gouttebarge et al., 2019).

Analysis of physiological and psychophysiological fatigue-related parameters (CK, SPR, and SPDOMS) did not demonstrate significant differences between matches played after 3–4 days and 6–7 days individual intervals. Rampinini et al. (2011) analysing post-match fatigue in a professional football team, showed that players generally recovered physically in 48 hours, which might explain the present findings. Although no differences were found for physiological parameters, the aforementioned high injury incidence might nevertheless suggest that the players were not completely recovered physically when playing long congested sequences of matches. This could be in part explained by different response rates for physiological and biomechanical load-adaptation pathways where these might occur over a different time frame and should subsequently be considered separately (Vanrenterghem et al., 2017).

In the present study, players did not present changes in physical performance in matches after an individual 6–7 day versus 3–4-day interval. No differences were found for the total distance covered in the matches (ES = −0.22) nor for the distances covered in different speed zones (14.4 to 20.0 km/ h: ES = −0.34; 20.1 to 25.0 km/h: ES = – 0.08; above 25.0 km/ h: ES = −0.19) when comparing matches played at 6–7 vs 3–4-day intervals. A reasonable explanation previously advanced may be that players utilize a pacing strategy whereby they consciously reduce the volume of low-intensity activity to facilitate the maintenance of key high-velocity movements (Jones et al., 2019)

This study is not without limitations. These include the non-inclusion of other constructs such as technical and tactical performance, the small sample size, and data from only one competitive season. Nevertheless, this study has relevant practical implications regarding congested calendar scenarios commonly occurring in top-level football. Future research should employ a more holistic approach including collective and individual measurements and additional measures of performance within a multiple club study.

Conclusion

High player availability and match participation rates were observed suggesting that players were substantially exposed to match-play during the long period of fixture congestion. The present findings reinforce the higher risk of muscle injury incidence during a CFP. Positively, no differences were reported in physical performance, or in fatigue parameters according to the time interval between matches (3–4 or 6–7-days). A shorter interval between matches, within the same extended congested fixture period, did not impact negatively upon physical performance or generate greater fatigue assessed using physiological and psychophysiological metrics. However, given the higher muscle injury incidence, the game's governing bodies should where possible avoid exposing players to long CFP in the sports calendar.

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